

RETAIL BUSINESS PERFORMANCE & PROFITABILITY ANALYSIS

Introduction

- The retail industry generates substantial volumes of transactional and inventory data, providing valuable opportunities for data-driven decision-making. Effective analysis of this data enables organizations to evaluate financial performance, identify profitable and underperforming products, optimize inventory levels, and understand changing customer and seasonal demand patterns.
- This project, Retail Business Performance & Profitability Analysis, focuses on analysing the Indian FMCG Retail Sales Customer Inventory (2024) dataset, which consists of 100,000 transaction records and 21 attributes. The dataset contains information related to sales transactions, product categories, brands, cities, sales channels, costs, revenue, profit margins, stock levels, reorder levels, and lead times.
- The primary objective is to transform raw transactional data into actionable business insights that can support profitability improvement, inventory optimization, product-level decision-making, and strategic planning.

Abstract

- This project presents an end-to-end retail analytics solution developed using Python, MySQL, and Microsoft Power BI. The analysis begins with data understanding and preprocessing, where the dataset is examined for data quality issues, missing values, duplicates, and inconsistencies. Feature engineering is subsequently performed to derive additional analytical attributes such as Month, Quarter, Year, Season, and Inventory Days.
- MySQL is used to store and analyse the prepared dataset through structured SQL queries. Key business metrics including revenue, cost, profit, profit margin, units sold, category profitability, brand performance, city-wise performance, channel performance, seasonal trends, and inventory levels are evaluated.
- Python, using Pandas and visualization libraries, is used for exploratory data analysis and to examine the relationship between inventory days and profitability. Finally, an interactive Power BI dashboard presents the key findings through KPI cards, charts, tables, scatter plots, and interactive filters.
- The resulting analysis provides a consolidated view of retail performance and helps identify profit-draining categories, low-performing products, slow-moving and overstocked inventory, and seasonal business opportunities.

Tools Used

- Python: Used for data cleaning, preprocessing, feature engineering, exploratory data analysis, and correlation analysis.
- Pandas & NumPy: Used for data manipulation, transformation, aggregation, and numerical analysis.
- Matplotlib & Seaborn: Used to create visualizations and identify patterns and relationships in the data.
- MySQL: Used for database creation, data storage, validation, and SQL-based business analysis.

- Microsoft Power BI: Used to develop the interactive dashboard, KPI cards, charts, tables, scatter plots, and filters.
- Dataset: *Indian FMCG Retail Sales Customer Inventory (2024)* containing 100,000 transaction records and 21 original attributes.

Steps Involved in Building the Project

The project follows a structured Data Preparation → Analysis → Visualization → Insight Generation methodology.

The FMCG retail dataset was extracted from the source ZIP file and loaded into Python. Its structure, dimensions, attributes, data types, statistical summaries, and data-quality characteristics were examined.

The dataset was checked for missing values, duplicate records, invalid values, and inconsistent data types. Missing customer-related values were appropriately treated, while date fields were converted into a standardized format. Data validation was performed to ensure the reliability of subsequent analysis.

Additional analytical attributes such as Month, Month Name, Quarter, Year, and Season were derived from the transaction date. An Inventory Days metric was developed using stock-on-hand and sales velocity to support inventory efficiency analysis.

The cleaned data was imported into MySQL for structured analysis. SQL queries were developed to evaluate overall KPIs, category and brand profitability, city performance, sales channels, store formats, seasonal performance, inventory levels, and reorder-level conditions.

Python was used to identify patterns and relationships within the data. Visual analysis was performed using Pandas, Matplotlib, and Seaborn. A correlation analysis between Inventory Days and Profitability was conducted to assess their statistical relationship.

The analysed data was integrated into Microsoft Power BI. An interactive dashboard was developed with Revenue, Profit, Profit Margin, Units Sold, Stock on Hand, and Inventory Days as key performance indicators. Additional visualizations were created for category, brand, city, channel, seasonal, and inventory performance.

The results were interpreted from a business perspective to identify high- and low-performing categories, profit-draining products, slow-moving inventory, potential overstock, and seasonal opportunities. Strategic recommendations were formulated to improve inventory efficiency and overall profitability.

Conclusion

The Retail Business Performance & Profitability Analysis project demonstrates the effective application of data analytics techniques to a real-world retail dataset. By integrating Python for data preparation and exploratory analysis, MySQL for structured business analysis, and Power BI for interactive visualization, the project establishes a complete analytical workflow from raw data to actionable insights.

The analysis enables stakeholders to evaluate sales performance, profitability, inventory efficiency, product behaviour, geographical performance, and seasonal trends from a unified perspective. In particular, the identification of low-margin products, slow-moving inventory, potential overstock situations, and seasonal demand patterns can support more informed operational and strategic decisions.

Overall, the project highlights the value of data-driven decision-making in retail management and provides a practical framework for improving profitability, inventory utilization, and business performance through systematic data analysis.